WAFIY ALMAS TSAMAR

Jambi, Indonesia | P: +62 821 7930 8645 | wafiyalmas2018@gmail.com | linkedin.com/in/wafiyalmas | portofolio

SUMMARY

Nanotechnology Engineering, Universitas Airlangga (GPA 3.52), with research experience in **semiconductor materials**, **hydrogen** technology, **lithium-ion batteries (NMC, LCO, LMO, LFP)** and battery recycling using **hydrometallurgy & pyrometallurgy**. Skilled in lab testing, process evaluation, and data analysis for quality and root-cause identification. Familiar with **OEE, SOPs,** and **BFD/PFD/P&ID reading** (certified). Demonstrated leadership as Head of Scientific Department and Chairperson of a university-level event. Served as an organizing committee member for three Focus Group Discussions (FGDs) on hydrogen technology at BRIN, involving the Ministry of Energy and Mineral Resources (ESDM EBTKE) and representatives from 100+ energy companies. A detail-oriented, analytical team player with leadership experience, eager to learn and contribute to process engineering and process improvement.

EDUCATION

Bachelor of Engineering (S.T.) - Nanotechnology Engineering | Universitas Airlangga *Aug 2022 - Jul 2026*

- Faculty of Advanced Technology and Multidisciplinary; achieved GPA: 3.52/4.0.

WORK EXPERIENCE

Badan Riset dan Inovasi Nasional (BRIN) | South Tangerang, Indonesia

Research Intern Merdeka Belajar Kampus Merdeka (MBKM) & Event Committee Member *Aug 2025 - Jul 2026*

- Conducted research activities related to hydrogen technology, fuel cells, electrolyzers, and catalytic materials.
- Served as an organizing committee member for three hydrogen technology focus group discussions (FGDs) activities with Kementerian ESDM EBTKE and directors from 100+ companies.
- Served as exhibitor/booth staff representing BRIN and Indonesia Fuel Cell and Hydrogen Energy (IFHE) at an event at ICE BSD, Tangerang.
- Developed experience in experimental planning, laboratory work, data interpretation, and scientific communication.

Amrillah Renewable Energy Research Group (AmrilLab) | Surabaya, Indonesia

Research Assistant *Sep 2024 - Apr 2025*

- Synthesized and characterized semiconductor nanomaterials for thin-film solar cells and photocatalysis, including CZTS, Ni-doped CZTS, BiOBr, and Al-doped ZnO.
- Evaluated solar cell performance using a solar simulator, including PCE, FF, Voc, and J-V curve analysis.
- Supported material development through systematic testing, data analysis, and research documentation.
- Co-authored the review article “Kirigami and Origami-Inspired Structures for (Flexible) Batteries and Supercapacitors” (manuscript under review).
- Co-authored the review article “Crafting a Solar Cell Device Integrated Textile” (manuscript under review).

ORGANIZATIONAL EXPERIENCE

Airlangga Student Research Group of Nanotechnology (ARGON)

Developed ARGON & Researcher at sensors & electronics division *Feb 2025 - Dec 2025*

- Developed ARGON’s organizational grand design and served as a researcher (sensors & electronics division)
- Contributed to electronics and sensor-related activities within the student research group.
- Concept Developer | Led conceptual design and coordination of a hydrogen-based electrolyzer system.
- Coordinated project development within the student research group.

Nanotechnology Engineering Student Association (HIMANO)

Head of Scientific Department *Feb 2024 - Dec 2025*

- Led the scientific department and coordinated research- and academic-oriented activities.
- Worked with organizational members to plan and execute departmental programs.

Airlangga Nanotechnology Week as part of Airnology 3.0

Chairperson *Feb 2024 - Sep 2024*

- Led the organizing committee and coordinated cross-functional teams throughout event planning and execution.
- Oversaw event operations, sponsorship and funding strategies, and coordination across event functions.
- Contributed to drafting Memoranda of Understanding (MoUs) for event partnerships and coordinated collaboration with external stakeholders.
- Ensured effective collaboration and alignment to achieve event objectives.

Tutoring Class Program

Person in Charge (PIC) *May 2023 - Jun 2023*

- Coordinated the implementation of a tutoring class program.

Community Service & Outreach

- Belajar Bersama Komunitas (BBK) | Made II, Sambikerep, Surabaya | Dec 2024 - Jan 2025 (Taught mathematics to students and delivered basic economics education to local residents).
- Organic Waste to Renewable Energy Outreach | Sutorejo, Surabaya | Sep 2023 (Promoted zero-waste lifestyles through community education on organic waste conversion into biogas).
- BEM FTMM x ARGON Community Outreach | Gempol, Pasuruan | Jun 2025 (Delivered practical education on nanochitosan fertilizer production and formulation, from raw materials to a farmer ready product, supported by SDGs UNAIR funding).

ACHIEVEMENTS

- Finalist – "Kompetisi Ilmiah Mahasiswa (KIM) 2025, Universitas Airlangga",
- Incentive Recipient – "Program Kreativitas Mahasiswa (PKM-GFT) 2024"
- Finalist – "International Youth Summit 2024, Kuala Lumpur, Malaysia".

CERTIFICATIONS & TRAINING

- ISO/IEC 17025 - Laboratory Management Systems.
- ISO 9001:2026 - Quality Management System.
- Unit Operations in Chemical Engineering.
- Basic Occupational Safety and Health (K3) Awareness.
- Job Safety Analysis (JSA) dan Hazard Identification, Risk assessment and Determining Control (HIRADC) Awareness,
- Awareness Personal Protective Equipment (PPE/APD) Awareness,
- How to Read P&ID, PFD & BFD used in Process Plant
- Supply Chain- MPS-RCCP-MRP-DRP-BOM
- Measuring Manufacturing Effectiveness-OEE, OOE, and TEEP
- English Language Proficiency Test (ELPT),

ADDITIONAL

Data & Software: Microsoft Excel, OriginPro, Gaussian, DWISM, BIOVIA Material Studio, HyperChem, ImageJ, Python, C++, SOLIDWORKS.

Technical Knowledge: Lithium-ion Battery Technology, Battery Recycling, Hydrometallurgy, Pyrometallurgy, Hydrogen Technology and Solar Cells.

Professional Skills: Teamwork, Communication, Public Speaking, Time Management, Analytical Thinking, Problem Solving, project Coordination, Research & Development, Scientific Writing, High Learning Motivation, Leadership.